

ucts for the thick-film technologies.

Screen-printing. Screen-printing is the process of pushing ink through a pattern woven on a mesh screen or stencil using a squeegee.

Drying/curing. After allowing a period of time after screen-printing for settling of the ink to occur, each layer of ink that is deposited is usually dried at a moderately high temperature (50° to 200°C) to evaporate the liquid component of the ink and fix the layer temporarily in position on the substrate so that it can be handled or stored before final processing. Some inks also require curing by exposure to UV light.

Firing. For many of the metal, ceramic and glass inks used in thick-film processes, a high-temperature (usually greater than 300°C) firing is required to fix the layers in position permanently on the substrate.

Separation of elements. This step is often necessary because many components are produced on one substrate at the same time. Thus, some means of separating the components from each other is required. This step may be achieved by wafer dicing.

Integration of devices. At this stage, the devices may require integration with other electronic components, usually in the form of a printed circuit board. This may be achieved by wire bonding or soldering.

Q2. What are the different types of stabilisers? What's the difference between dynamic and static voltage stabilisers?

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A2. Stabilisers stabilise the voltage automatically to protect the sensitive equipment from under-voltage, over-voltage, surge voltage, line disturbance and lightning. There are many types of voltage stabiliser systems. However, these can be broadly categorised into ferro-resonance, servo motor control and automatic tap-change types. They can be further divided into dynamic and static systems.

Servo motor control systems consist

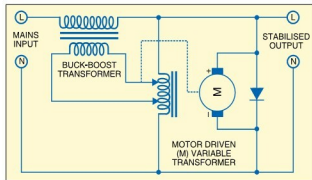


Fig. 1: Circuit of servo voltage stabiliser

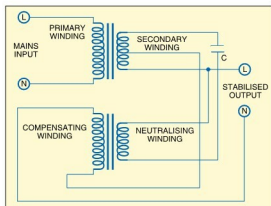


Fig. 2: Circuit of ferro-resonant voltage stabiliser

of a transformer having its secondary winding connected between the mains supply and the load. The primary voltage is automatically controlled through a motor-driven variable transformer, ensuring a continuous, smooth and very stable output voltage.

To reduce the spikes to totally harmless levels, it is often necessary to ask your supplier to fit additional spike attenuation protection. While electronic servo stabilisers do contain some moving parts, these have proved very reliable over some of the most demanding power conditions with only a low level of maintenance.

Ferro-resonance voltage stabilisers use a saturable transformer winding in series with a linear choke, a compensating winding connected in series with an output circuit and a capacitor making a ferro-resonant circuit with the transformer winding, in which an additional compensating winding is provided in series with the capacitor to compensate variations of voltage in

the capacitor circuit, thereby giving a well regulated and stabilised output voltage.

With no moving parts and no electronic control circuitry, there is no need for maintenance and the system is virtually an install-and-forget solution. The design can withstand high

instantaneous overloads and suppress lightning-induced spikes and surges.

Dynamic voltage stabiliser. Fast voltage collapse is a new phenomenon in today's stressed power systems. To prevent fast voltage collapse, sufficient amount of compensation is needed. A power system voltage stabiliser allows for dynamic voltage stability enhancement of power systems and/or prevention of fast voltage collapse. It is implemented through coordinated control of dynamic reactive sources.

Static stabiliser. It is an SMPS-type voltage stabiliser for mains voltage (AC input and AC output). The power stage is an IGBT chopper control. The chopping frequency is around 20 kHz, which ensures absolutely silent operation and pure sine wave output without waveform distortion.

The control section is based on a dsPIC controller that ensures quick correction of the output, which is not possible in conventional relay-type or servo-controlled stabilisers. A liquid crystal display shows all parameters like input voltage, output voltage and connected load.

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